

Tutorial – 01

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Questions

1. Briefly explain the need of a programming language?

To communicate with each other language is a must to every one so to communicate with computer we can't use other languages we need to use programming language to communicate with it . By using programming language can develop applications to perform specific task.

2. Compare and contrast the differences between followings;

- a. Source Code vs. Machine Code

Source Code	Machine Code
• The program that programmer writes called as a source code. • Source Code is easy to understand • Instruction are written by using English words. • Easy to modify .	• The code that can understand by the machine called as machine code. • Machine code is hard to understand. • Instructions are written in binary digits . • Difficult to modify.

b. High Level Language vs. Low Level Language

High Level Language	Low Level Language
• Less memory capacity • This is a programming friendly language. • Easy to read ,write and debug codes. • Performance is poor	• More memory capacity • This is a machine friendly language. • Difficult to write and debug the codes. • Have a higher performance.

c. Compiler vs. Interpreter

Compiler	Interpreter
• A compiler converts the source code into machine code at once. • More efficient . • Difficult to detect the errors • It needs less time	• Interpreter converts the source code into machine code using line by line. • Less efficient. • Compare with the compiler it is easy to detect errors in Interpreter. • It needs more time

d. Structured Language vs. Object Oriented Language

Structured Language	Object Oriented Language
• This program represents as a logical structure • Difficult to modify. • Data is not secured • Less Flexibility	• This program is written as a collection of objects that communicate with each other. • Easy to Modify. • Data is secured

e. C vs. C++

C	C++
• C is a procedural language • Top down approach is used • For input scanf() function is used and for output printf() function is used. • No inheritance in C .	• C++ is an object oriented programming language . • Bottom up approach is used . • For input cin>> function is used and for output cout<< function is used. • Can have Inheritance in C++.

f. C++ v. Java

C++	Java
• Support for pointer concept. • Object oriented programming . • Based on C language • C++ is a dependent language .	• Doesn't support for pointer concept. • Pure object oriented concept . • Based on c and c++. • Java is a Independent Language

g. Syntax error vs. Logical error

Syntax Error	Logical Error
• This error happens because of grammatical mistakes. • It is easy to identify the error. • Compiler show the error when occurs and also the location.	This will occur because of a fault in algorithm . • Very difficult to identify the error . • Programmer should identify the error by own.